

Kubernetes Cluster

Kubelet Certificate Expiration

Troubleshooting & Recovery Runbook

Kubernetes v1.32.8 • Ubuntu 24.04 LTS • containerd 2.0.6

Environment

Node	Role	IP
k8s-master	Control Plane	10.0.2.14
k8s-worker1	Worker	10.0.2.12
k8s-worker2	Worker	10.0.2.16

1. Control Plane Recovery — k8s-master

1.1 Check certificate expiration

Node: k8s-master

```
sudo kubeadm certs check-expiration
```

Purpose: Identify expired Kubernetes control-plane certificates.

1.2 Create backups

Node: k8s-master

```
sudo cp -a /etc/kubernetes/kubelet.conf /etc/kubernetes/kubelet.conf.backup-20260929
sudo cp -a /var/lib/kubelet/pki /var/lib/kubelet/pki.backup-20260929
```

Purpose: Preserve kubelet configuration and certificates before changes.

1.3 Renew Kubernetes certificates

Node: k8s-master

```
sudo kubeadm certs renew all
```

Purpose: Renew expired Kubernetes control-plane certificates.

1.4 Preserve the old kubelet configuration

Node: k8s-master

```
sudo mv /etc/kubernetes/kubelet.conf /etc/kubernetes/kubelet.conf.expired-20260929
```

Purpose: Keep the expired configuration as a backup.

1.5 Regenerate master kubelet kubeconfig

Node: k8s-master

```
sudo kubeadm init phase kubeconfig kubelet --node-name k8s-master --kubeconfig-dir /etc/kubernetes --cert-dir /etc/kubernetes/pki
```

Purpose: Generate a new kubelet kubeconfig using valid certificates.

1.6 Restart kubelet

Node: k8s-master

```
sudo systemctl restart kubelet
sudo systemctl status kubelet --no-pager -l
```

Purpose: Load the renewed credentials and verify kubelet is running.

1.7 Verify nodes

Node: k8s-master

```
sudo kubectl --kubeconfig=/etc/kubernetes/admin.conf get nodes -o wide
```

Purpose: Confirm the control plane is Ready and identify affected workers.

2. Worker Recovery — k8s-worker1 / k8s-worker2

Root cause found: expired kubelet authentication material and missing bootstrap-kubelet.conf. Existing worker nodes were repaired without kubeadm reset or rejoin.

k8s-worker1

2.1 Generate a new worker kubeconfig

Node: k8s-master

```
sudo kubeadm kubeconfig user --org system:nodes --client-name system:node:k8s-worker1 > /tmp/kubelet-k8s-worker1-new.conf
sudo chmod 600 /tmp/kubelet-k8s-worker1-new.conf
```

Purpose: Generate a fresh kubeconfig for the existing worker using the cluster CA.

2.2 Copy the kubeconfig to the worker

Node: k8s-master

```
sudo scp /tmp/kubelet-k8s-worker1-new.conf esidi92@10.0.2.12:/tmp/kubelet-k8s-worker1-new.conf
```

Purpose: Transfer the new kubelet authentication configuration.

2.3 Backup existing kubelet.conf

Node: k8s-worker1

```
sudo cp -a /etc/kubernetes/kubelet.conf /etc/kubernetes/kubelet.conf.backup-20260929
```

Purpose: Preserve the existing worker configuration.

2.4 Install the new kubelet.conf

Node: k8s-worker1

```
sudo cp -a /tmp/kubelet-k8s-worker1-new.conf /etc/kubernetes/kubelet.conf
sudo chown root:root /etc/kubernetes/kubelet.conf
sudo chmod 600 /etc/kubernetes/kubelet.conf
```

Purpose: Replace expired kubelet credentials securely.

2.5 Restart kubelet

Node: k8s-worker1

```
sudo systemctl restart kubelet
sudo systemctl status kubelet --no-pager -l
```

Purpose: Reload credentials and verify kubelet is active.

2.6 Verify worker registration

Node: k8s-master

```
sudo kubectl --kubeconfig=/etc/kubernetes/admin.conf get nodes -o wide
```

Purpose: Confirm the worker becomes Ready.

k8s-worker2

2.1 Generate a new worker kubeconfig

Node: k8s-master

```
sudo kubeadm kubeconfig user --org system:nodes --client-name system:node:k8s-worker2 > /tmp/kubelet-k8s-worker2-new.conf
sudo chmod 600 /tmp/kubelet-k8s-worker2-new.conf
```

Purpose: Generate a fresh kubeconfig for the existing worker using the cluster CA.

2.2 Copy the kubeconfig to the worker

Node: k8s-master

```
sudo scp /tmp/kubelet-k8s-worker2-new.conf esidi92@10.0.2.16:/tmp/kubelet-k8s-worker2-new.conf
```

Purpose: Transfer the new kubelet authentication configuration.

2.3 Backup existing kubelet.conf

Node: k8s-worker2

```
sudo cp -a /etc/kubernetes/kubelet.conf /etc/kubernetes/kubelet.conf.backup-20260929
```

Purpose: Preserve the existing worker configuration.

2.4 Install the new kubelet.conf

Node: k8s-worker2

```
sudo cp -a /tmp/kubelet-k8s-worker2-new.conf /etc/kubernetes/kubelet.conf
sudo chown root:root /etc/kubernetes/kubelet.conf
sudo chmod 600 /etc/kubernetes/kubelet.conf
```

Purpose: Replace expired kubelet credentials securely.

2.5 Restart kubelet

Node: k8s-worker2

```
sudo systemctl restart kubelet
sudo systemctl status kubelet --no-pager -l
```

Purpose: Reload credentials and verify kubelet is active.

2.6 Verify worker registration

Node: k8s-master

```
sudo kubectl --kubeconfig=/etc/kubernetes/admin.conf get nodes -o wide
```

Purpose: Confirm the worker becomes Ready.

3. Configure kubectl for the Normal User — k8s-master

3.1 Configure user kubeconfig

Node: k8s-master

```
mkdir -p ~/.kube
sudo cp /etc/kubernetes/admin.conf ~/.kube/config
sudo chown $(id -u):$(id -g) ~/.kube/config
```

Purpose: Allow kubectl to work without specifying --kubeconfig every time.

3.2 Final workload verification

Node: k8s-master

```
kubectl get pods -A
```

Purpose: Verify workloads across all namespaces are accessible.

4. Final Expected State

```
kubectl get nodes -o wide
```

k8s-master	Ready	control-plane
k8s-worker1	Ready	worker
k8s-worker2	Ready	worker

Result: All three nodes were restored to Ready without resetting or rejoining the existing cluster.

5. Commands Intentionally NOT Used

```
kubeadm reset
rm -rf /var/lib/kubelet/pki
rm -rf /etc/kubernetes/pki
kubectl delete node <node>
```

Reason: the cluster CA, Node objects, and existing cluster membership remained valid. The issue was expired kubelet authentication, so destructive rebuild/rejoin operations were unnecessary.

6. Quick Recovery Flow

Check cluster → Check certificate expiration

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Renew control-plane certificates

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Regenerate master kubelet.conf

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Restart master kubelet

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Identify NotReady worker

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Generate worker kubeconfig on master

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Copy kubeconfig to worker

↓

Backup + replace worker kubelet.conf

↓

Restart worker kubelet

↓

Verify worker = Ready

↓

Repeat for other affected workers

↓

kubectl get nodes

kubectl get pods -A

7. Operational Notes

- Never paste embedded private keys or certificate data into chat or tickets.
- Do not run `kubeadm reset` unless a full node rejoin is intentionally planned.
- Always back up `kubelet.conf` before replacing it.
- Use the healthy control plane to generate a worker kubeconfig when the worker cannot bootstrap itself.
- After recovery, verify both node health and workload visibility.